

BT-302 OBJECTIVE MID TERM-

100% GUESS PAPER 2024

IMMUNOLOGY BY

SULMAN ALI GUJJAR (ZOOLOGIST)

0341-7547044

OBJECTIVES

- ❖ Another term used for the physical barriers of innate immune system.....(**Anatomical Barriers**)
- ❖ Which of the following are primary lymphoid organ.... (**Bone Marrow, Thymus**)
- ❖ Which secondary lymphoid organ is responsible for the filtration of blood..... (**Spleen**)
- ❖ (**Phagocytosis**)..... The process of engulfing the invading or infectious agent by phagocytes.
- ❖ A substance that is non-immunogenic but can react with products of specific immune response.....(**Hapten**)
- ❖ Which antibodies are complement fixing antibody.....(**IgM and IgG**)
- ❖ IgG exist in the form of.....(**Monomer**)
- ❖ Strength of binding of antigen with many antigenic determinant & multivalent antibody.....(**Avidity**)
- ❖ Agglutinate is term used for visible clumps formation in.....(**Agglutination**)
- ❖ The process by which a B-Cell changes the class of Ab but not specificity.....(**Class Switching**)
- ❖ Antibody from a single antibody producing B-cell.....(**Monoclonal Ab**)
- ❖ hybridoma cell can grow on **HAT medium**
- ❖ Snake venom pathway **Both pathways**
- ❖ TCR have **carbohydrates moiety**
- ❖ Immune response is carried out by **carrier immunogenic**
- ❖ **IgM** is the **third most common** serum Ig
- ❖ **IgA** is the **2nd most common** serum Ig
- ❖ **IgG** is the only class of Ig that crosses the **placenta**
- ❖ **Complement fixation** tests are most commonly used to assay for antibody in a test sample but they can be modified to measure antigen
- ❖ **Sandwich ELISAs** are the **most common** type of ELISA
- ❖ **Competitive ELISAs** are commonly used for **small molecules**
- ❖ **IgA deficiency**: This is the **commonest form** of immunoglobulin deficiency
- ❖ The CD3 stands for **Cluster of differentiation 3**
- ❖ **Age** is another biological factor which can also influence **immunogenicity**
- ❖ **Diabetes mellitus** is **non-infectious** disease
- ❖ IgA is the major class of Ig in secretions - **tears, saliva, colostrum, mucus**

- ❖ **Agglutination Reactions:** These are antigen and antibody reactions in which the particulate nature of antigen causes the reaction with its specific antibody to form visible clumps. Such visible clumps are called as **agglutinates**.
- ❖ **Antibodies** are produced by **lymphocytes**
- ❖ Natural killer (NK) cells (May develop from both **Myeloid and Lymphoid progenitor**).
- ❖ NK cells are produced from the **bone marrow** and are found in the bloodstream
- ❖ Which antigen in anaphylaxis...**IgE**
- ❖ Humans can make 10^7 - 10^8 different Ab molecules.
- ❖ Recombination involves rearrangement of DNA in somatic cells... **Somatic recombination**.
- ❖ Lymphoid linkage progenitor cells give rise to ----- B- lymphocytes, T- Lymphocytes, natural Killer, **All of the given**.
- ❖ Major class of antibody produced during primary antibody response is ----- **IgM**
- ❖ During complement fixation test, lysis of RBCs indicates a ----- Complement fixation test. **Negative**
- ❖ Which of the following comes under mechanical factors regarding anatomical barriers? Skin, Epithelium membrane, Desquamation, **All of the given**
- ❖ Lymph nodes, spleen and Peyer's patches are. **Secondary lymphoid organs**
- ❖ Substance that is non-immunogenic but can react with products of specific immune response called..... **Heptan**
- ❖ Humans can make 10^7 to 10^8 different Ab molecules
- ❖ T-cells maturation in... **Thymus**
- ❖ Ag & Ab reaction which can be detected on the basis of substrate utilization by enzyme ...**ELISA**
- ❖ Nature of Ag & Ab reactions... **Non covalent bonding** Immunology is the study ofboth **immune system & immune response**
- ❖ Lag phase of secondary exposure is... **Shorter as compare to primary exposure**
- ❖ Peristalsis movement occurs in **digestive tract**
- ❖ **Valency** is the Number of antigenic determinants which can bind with **immunoglobulin**.
- ❖ Ag is presented to T-cells with..... **Class II MHC**
- ❖ Suppose 10,000 lymphocytes recirculate every hour. If 100 of these encounter an antigen in lymph node then what percentage it achieved... **1%**.
- ❖ Antigen selects those clones of cells that have the **appropriate receptor**.
- ❖ Suppose 10,000 lymphocytes recirculate every hour. If 100 of these encounter an antigen in the lymph node. What percentage is activated? (**1%**)
- ❖ The expanded clones of B cells differentiate into **plasma cells**
- ❖ **Skin and mucus** membrane are the examples of **anatomical barriers** that provides immunity
- ❖ Examples of infectious diseases include the **common cold, COVID-19, SARS and tuberculosis**
- ❖ Which of the following disease is transmitted between individuals is **tuberculosis**
- ❖ The prospective donor & recipient are tested for compatibility prior to **transplantation**. That's why MHC is also called as **Human leucocyte antigen (HLA)**
- ❖ Agglutination e.g Anti-nuclear antibodies (ANA) for **SLE & RA**
- ❖ The antibodies which render such kind of reactions are called as “**agglutinins**”.
- ❖ **IgD** is primarily found on **B cell surfaces**
- ❖ Red blood cell surface antigens are **glycoproteins**
- ❖ These antigens are **protein** in nature **T-dependent antigens**
- ❖ Antigens are primarily composed of **proteins**
- ❖ penicillin are **haptens**
- ❖ **Chemical coupling** of a hapten to a large protein, called a **carrier**, yields an immunogenic hapten-carrier conjugate
- ❖ Antibodies are produced by **lymphocytes**, which are formed in lymph nodes
- ❖ **Naïve (virgin) lymphocytes** enter the lymph nodes from the blood via **High Endothelial Venules (HEVs)**
- ❖ Tertiary lymphoid tissues, which normally contain fewer lymphoid cells than secondary lymphoid organs, can import lymphoid cells during an inflammatory response. Most

prominent of these are **cutaneous-associated lymphoid tissues**.

- ❖ **Enzyme Linked Immunosorbent Assay (ELISA)**
- ❖ Less specific component of immune system **innate immunity**.
- ❖ **Immunity** Is the study of state of protection from infectious disease.
- ❖ Immune system is the **Defense system** that has evolve to protect animals from invading pathogenic microorganisms.
- ❖ Example of infectious disease include the **All(hepatitis, TB, COVID)**.
- ❖ NON infectious disease cause such a fact as **All**.
- ❖ Risk of contacting Type 2 diabetes can be decreased by maintaining a healthy lifestyle **True**.
- ❖ Infectious disease are transmitted from person to person through the transfer of pathogen such **as All (bacteria, fungi, virus)**
- ❖ THE IMMUNE SYSTEM MALFUNCTION by losing it sense of self and non self which permits an immune attack upon the host is called **autoimmunity**.
- ❖ Transplantation studied have major role in the development of **immunodeficiency**
- ❖ **AIDS** THIS RESULTS FROM infection with the HIV.
- ❖ An additional danger in transplantation is that any transplanted cells with immune function may view the new host **as non self and react against it**. This reaction, which is termed graft-versus-host disease,
- ❖ Innate immune system **is uniform** in all members of a species,
- ❖ **Antibodies** can distinguish between two protein molecules that differ in only a single amino acid.
- ❖ The epidermis contains several layers of tightly packed **epithelial cells**.
- ❖ Tissue damage caused by a wound or by an invading pathogenic microorganism induces a complex sequence of events collectively known as the **inflammatory response**.
- ❖ The skin consists of two distinct layers: a thinner outer layer— thicker layer— the **epidermis, dermis**.
- ❖ *Bacillus anthracis* are killed by the higher body temperature of **birds (39°C)**.
- ❖ Interferon found in blood and lymph kills **viruses**.
- ❖ Skin provides **first line of defense** by preventing entry of microorganisms.
- ❖ Mammals are susceptible but birds are resistant to **anthrax**.
- ❖ TLR2 recognizes the **lip polysaccharide (LPS)** found on Gram-negative bacteria.
- ❖ **Fibrin** is a protein which seals wounds and stops bleeding.
- ❖ **Mast cells**- are granulocytes, that secrete heparin, histamine, other factors. They help in **wound healing, angiogenesis, and elimination of parasites**.
- ❖ BASOPHILL **histamine-rich** granules and involve in inflammatory responses.
- ❖ **Neutrophills** eliminate pathogens especially bacteria by phagocytosis.
- ❖ **Liver have speciall cells called kuffer cells**
- ❖ SOS signals stimulate endothelial cells near the site of the infection to express cell **adhesion molecules such as ICAM-1**
- ❖ NADPH oxidase components, which are involved in the generation of **toxic oxygen products**.
- ❖ Toll-like receptors results in phagocytosis and the release of **inflammatory cytokines**
- ❖ Bacteria with **IgG antibody** on their surface have the Fc region exposed
- ❖ Scavenger receptors bind a wide variety **of polyanions** on bacterial surfaces resulting in phagocytosis of bacteria.
- ❖ **The Alternative Pathway Is Antibody-Independent**
- ❖ Carbohydrate composition and its sialic acid content on the cell surface play an important role in the activation of the **alternate pathway**.
- ❖ alternative pathway is initiated in most cases by **cell-surface** constituents that are foreign to the host
- ❖ **Tickover** occurs spontaneously at a rate of ~1% of total **C3 per hour**
- ❖ lectin pathway is activated by the binding **of mannose-binding lectin (MBL)** to mannose residues on glycoproteins or carbohydrates on the surface of microorganisms.
- ❖ MBL is an acute phase protein produced **in inflammatory responses**.
- ❖ MASP-1 and -2 proteins have structural similarity to **C1r and C1s** and mimic their activities
- ❖ Lectins are proteins that recognize and bind to specific **carbohydrate targets**

- ❖) Complement proteins also served as chemoattractants for the movement of phagocytic cells towards the **site of infection**.
- ❖)

❖ C3 and Factor B	❖ Severe bacterial infections
❖ C3b-INa, C6 and C8	❖ Severe Neisseria infections
❖ Deficiencies of early C components C1, C4, C2.	❖ Systemic lupus erythematosus (SLE), glomerulonephritis and polymyositis
❖ C1-inhibitor	❖ Hereditary angioedema

- ❖ Complement may also be activated during reactions against incompatible **blood transfusions**,
- ❖ The **thymus and bone marrow** are the primary (or central) lymphoid organs
- ❖ The lymph nodes, spleen, and various **mucosal-associated lymphoid tissues (MALT)** such as **gut-associated lymphoid tissue (GALT)** are the secondary (or peripheral) lymphoid organs,
- ❖ B cells originate **in bone marrow**.
- ❖ **stromal cells** within the bone marrow interact directly with the B cells
- ❖ **T cells arise** in the thymus, and in many mammals—humans and mice for example
- ❖ **MALT constitutes about 50%** of the lymphoid tissue in human body.
- ❖ the antigens trapped in the lymph nodes are responsible for the activation of **lymphocytes** present there and cause the immune response.
- ❖ **The lymphoid lineage**
- ❖ T lymphocytes (T cells)
- ❖ B lymphocytes (B cells)
- ❖ Natural killer cells (NK cells)
- ❖ **The myeloid lineage**
- ❖ Monocytes, macrophages
- ❖ Langerhans cells, dendritic cells
- ❖ Megakaryocytes
- ❖ Granulocytes (eosinophils, neutrophils, basophils)
- ❖ a **multipotent stem cell** differentiates along one of two pathways, giving rise to either a common lymphoid progenitor cell or a common myeloid progenitor cell.
- ❖ lymphoid progenitor cells give rise to **B, T, and NK (natural killer)** cells and some dendritic cells
- ❖ Stem cells THAT HAVE FOLLOWING PROPERTIES **differentiate into other cell types; they are self-renewing**
- ❖ All blood cells arise from a type of cell called **the hematopoietic stem cell (HSC)**
- ❖ platelet-generating cell are called **(the megakaryocyte)**.
- ❖ Erythrocytes are **non-nucleated cells**,
- ❖ Naive (virgin) lymphocytes enter the lymph nodes from the blood via **High Endothelial Venules (HEVs)**
- ❖ lymphatics and return to the blood via the **thoracic duct**.
- ❖ It is estimated that **1-2%** of lymphocytes recirculate every hour.
- ❖ the effector cells can leave the lymph nodes via the lymphatics and return to the blood via the **thoracic duct**.
- ❖ the chances for a successful encounter are greatly **enhanced** by the recirculation of lymphocytes
- ❖ the number of lymphocytes specific for a given antigen is **greatly amplified**
- ❖ The amplified population of **memory cells** accounts for the rapidity and intensity that distinguishes a secondary response from the primary response.
- ❖ the antigenic specificity of each of these clones is determined by the specificity of **the antigen-binding receptor** on the membrane of the clone's lymphocytes.
- ❖ Infection by extracellular bacteria induces production of **humoral antibodies**,
- ❖ opsonin that increases phagocytosis and thus clearance of the bacterium.
- ❖ Extracellular bacteria can be pathogenic because they induce a localized inflammatory response or because they produce **toxins**.
- ❖ the toxin produced by diphtheria, which exerts a toxic effect on the cell by **blocking protein synthesis**.

- ❖ Endotoxins, such as lipopolysaccharides (LPS), are generally components of **bacterial cell walls**,
- ❖ primary defense against pathogens in the cytosol is the **cytotoxic T lymphocyte** (Tc or CTL).
- ❖ Th cells are a subset of T cells that express a unique antigen on their surface **called CD4**
- ❖ the primary defense against a pathogen within vesicles is a subset of helper **T lymphocytes** (Th1).
- ❖ Th1 cells are a subset of T cells that express a unique antigen on their surface **called CD4**.
- ❖ Because **antibodies do not get into host cells**, they are ineffective against intracellular pathogens
- ❖ **Hapten** is a molecule that reacts with specific antibody but is not immunogenic by itself, it can be made immunogenic by conjugation to a suitable carrier.
- ❖ An epitope, also known as antigenic determinant, is the part of an antigen that is recognized by an antibody, **B-cell receptor or T-cell receptor**.
- ❖ A specific protein which is generated in response to an immunogen and which reacts with **an antigen is called Antibody**
- ❖ Immunogenicity is determined, by **four** properties of the immunogen:
- ❖ The development of both humoral and cell-mediated immune responses requires interaction of T cells with antigen that has been processed and presented together with **MHC molecules**.
- ❖ The most active immunogens tend to have a molecular mass of **100,000 daltons (Da)**.
- ❖ copolymers composed of different amino acids or sugars are usually **more immunogenic** than homopolymers of their constituents.
- ❖ Age is another biological factor which can also influence immunogenicity **TRUE**
- ❖ **GENOTYPE OF THE RECIPIENT ANIMAL** Age properties that contribute to immunogenicity **of the Biological System**
- ❖ An insufficient dose will not stimulate an immune response either because it fails to **activate enough lymphocytes** .
- ❖ antigen administered subcutaneously moves first to **local lymph nodes**
- ❖ Antigen administered intravenously is carried first to the **spleen**.
- ❖ An antibody response is measured by determining the level of antibody present in the serum of immunized animals
- ❖ Nucleic acid is non immunogenic because simplicity of the **structure, molecular flexibility, and rapid degradation**. M
- ❖ polysaccharides are present in the form of **side chains** and give rise to antigenicity.
- ❖ **Lipids**: These are non-immunogenic except cardiolipin
- ❖ -independent antigens are antigens which can directly stimulate the B cells to produce antibody without the requirement for T cell help In general, polysaccharides are T-independent antigen
- ❖ Many of these antigens can activate B cell clones specific for other antigens (**polyclonal activation**)
- ❖ **Proteins** are T-dependent antigens.
- ❖ superantigens can activate large numbers of T cells irrespective of their **antigenic specificity**.
- ❖ The large number of T cells activated by these superantigens results in excessive production of **cytokines**
- ❖ toxic-shock syndrome toxin (TSST1) **from Staphylococcus aureus**
- ❖ haptens, small organic molecules that are antigenic but **not immunogenic**
- ❖ Animals immunized with such a conjugate produce antibodies specific for **,the hapten determinant,unaltered epitopes on the carrier protein ,new epitopes formed by combined parts of both the hapten and carrier.**
- ❖ the **lipopolysaccharide (LPS, also called endotoxin)** of **Gram-negative bacteria**;
- ❖ The valency of antibody refers to the number of **antigenic determinants** that an individual antibody molecule can bind.
- ❖ Some immunoglobulins also bind to receptors on placental trophoblasts, which results in transfer of the immunoglobulin across **the placenta**.
- ❖ Antibodies are antigen binding proteins present on **the B-cell membrane**.
- ❖ Antibody molecules have a common structure of **four peptide chains**.
- ❖ These are folded regions which contain an intra-**chain disulfide bond**. Usually, heavy chains contain four domains while light chains composed on two domains.
- ❖ Antibodies are glycoproteins; with few exceptions, the sites of attachment for carbohydrates are restricted to the **constant region**.

- ❖ Diversity in the Variable-Region Domain Is Concentrated in CDRs
- ❖ The **three** heavy-chain and **three** light-chain CDR regions are located on the loops that connect the strands of the VH and VL domains.
- ❖ **Variability = # of different amino acids at a given position / Frequency of the most common amino acid at given position TRUE**
- ❖ When the γ -globulin fraction of serum is separated into high- and low-molecular weight fractions, antibodies of around 150,000-MW, designated as **immunoglobulin G (IgG)** are found in the low molecular-weight fraction.
- ❖ Inside the phagocyte, the pathogen becomes the target of various destructive processes that include **enzymatic digestion**, oxidative damage, and the membrane-disrupting effects of antibacterial peptides
- ❖ DIGESTED IgG with pepsin
- ❖ The antibodies which cause immunoprecipitation reactions are called as **precipitins**.
- ❖ test will only work with complement fixing antibodies (**IgG and IgM are best**).
- ❖ If no antigen/antibody complexes are present in the tube, none of the complement will be fixed.
- ❖ Since only **one antibody is used** in a direct ELISA, they are less specific than a sandwich ELISA.
- ❖ Antibodies are produced by **lymphocytes**, which are formed in lymph nodes.
- ❖ **humoral immunity** depends on the B Cells while cell immunity depends on the T Cells.
- ❖ The T and B lymphocytes (T and B Cells) are involved in the **acquired or antigen-specific** immune response depends on the B Cells while cell immunity depends on the T Cells.
- ❖ **TISSUE** immune system is that it normally distinguishes between self and non-self and only reacts against non-self.
- ❖ A response to a particular antigen is specific for that antigen or a few closely related antigens.
- ❖ the host's cells and enzymes metabolize the antigen IN CATABOLIC DECAY PHASE
- ❖ Equilibrium phase the antigen equilibrates between the vascular and extravascular compartments by **diffusion**.
- ❖ primary response, the major class of antibody produced **is IgM**
- ❖ The antibodies that persist in the secondary response are the **IgG antibodies**.
- ❖ The duration of latent phase will vary depending on the antigen but it is usually 5 to 7 days.
- ❖ antibody synthesis is balanced by antibody decay so that there is no net increase in antibody concentration in **Plateau or steady-state phase**
- ❖ there is no memory response in case of **T-independent antigens**
- ❖ If the second polyA site is used, the pre-mRNA is processed to produce a membrane form of the **immunoglobulin**.
- ❖ If the first polyA site is used, the pre-mRNA is processed to produce a **secreted protein**.
- ❖ **introns** (intervening non-coding sequences)
- ❖ The immunoglobulin heavy and light chains are coded for **by three** separate gene families each one on a separate chromosome
- ❖ The phenomenon of enhanced rate of point mutation in the immunoglobulin **V region genes** is called as **Somatic hyper** mutation
- ❖ humans can make **10^7 - 10^8** different antibody molecules.
- ❖ activation induced cytidine deaminase (AID) which is considered essential in **DNA deamination** for somatic hyper mutation.
- ❖ the V, J and D exons, there are unique sequences referred to as recombination **signal sequences (RSS)**
- ❖ Mutations in the genes for these proteins results in a severe combined immunodeficiency disease (**both T and B cells are deficient**),
- ❖ Each RSS consists of a conserved nonamer and a conserved **heptamer**
- ❖ B-cells with higher affinity Ig can compete better for limited amount **of Ag**
- ❖ Following Ag activation, V regions of heavy & light chains are diversified by **somatic hypermutation**
- ❖ Affinity maturation occurs during secondary Ab response
- ❖ Affinity maturation binding affinity of BCR for its cognate ligand
- ❖ During an antibody response to a **T-dependent antigen** a switch occurs in the class of Ig produced from IgM to some other class (except IgD).
- ❖ the antibody specificity is determined by the **hypervariable regions** within the V region
- ❖ the path leading to the association of protein fragments with MHC molecules differs for class I and class **II MHC**.

- ❖ **MHC class II** molecules present fragments derived from extracellular (exogenous) proteins that are located in an intracellular compartment.
- ❖ **MHC class I** molecules present degradation products derived from intracellular (endogenous) proteins in the cytosol
- ❖ The **intracellular source of antigen** which is also called as **Endogenous antigen** reside
- ❖ Synthesis and assembly of class I heavy chain and beta2 microglobulin occurs in the **endoplasmic reticulum**.